# MINDQUEST 2026 - MASTER QUESTION BANK

Official Competition Format Specification Sample | 30 Standardized Items

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**Q1. What is the primary function of the Central Processing Unit (CPU) in a computer system?**

A) Long-term data archiving  B) Instruction execution and arithmetic logic
C) Digital-to-analog audio conversion  D) Power regulation across motherboard circuits

**Answer: B**

**Q2. Which data structure uses the First In, First Out (FIFO) principle for element access?**

A) Stack  B) Binary Search Tree
C) Queue  D) Max Heap

**Answer: C**

**Q3. What is the worst-case time complexity of standard QuickSort with poor pivot selection?**

A) O(n log n)  B) O(n^2)
C) O(n)  D) O(log n)

**Answer: B**

**Q4. In relational databases, which normal form eliminates transitive dependencies?**

A) First Normal Form (1NF)  B) Second Normal Form (2NF)
C) Third Normal Form (3NF)  D) Boyce-Codd Normal Form (BCNF)

**Answer: C**

**Q5. Which cryptographic protocol operates at the Transport Layer of the OSI model to secure internet traffic?**

A) IPsec  B) TLS (Transport Layer Security)
C) SSH  D) PGP

**Answer: B**

**Q6. What does the ACID acronym stand for in transactional database management?**

A) Atomicity, Consistency, Isolation, Durability  B) Accuracy, Concurrency, Indexing, Distribution
C) Asynchronous, Cached, Idempotent, Decoupled  D) Allocation, Clustering, Integrity, Duplication

**Answer: A**

**Q7. Which sorting algorithm is guaranteed to be stable and has O(n log n) worst-case time complexity?**

A) HeapSort  B) MergeSort
C) QuickSort  D) Selection Sort

**Answer: B**

**Q8. In modern operating systems, what is thrashing caused by?**

A) Corrupted file allocation tables  B) Excessive page swapping between RAM and secondary storage
C) CPU overheating leading to thermal throttling  D) Deadlocks in multi-threaded mutex acquisition

**Answer: B**

**Q9. Which HTTP status code signifies that a resource has been permanently moved to a new URI?**

A) 301 Moved Permanently  B) 302 Found
C) 304 Not Modified  D) 307 Temporary Redirect

**Answer: A**

**Q10. What is the principal benefit of using an immutable data structure in concurrent programming?**

A) Faster memory allocation on the stack  B) Elimination of race conditions without lock contention
C) Smaller footprint inside CPU cache lines  D) Automatic compilation to GPU shaders

**Answer: B**

**Q11. Which graph traversal algorithm uses a queue and finds the shortest path on unweighted graphs?**

A) Depth-First Search (DFS)  B) Breadth-First Search (BFS)
C) Floyd-Warshall Algorithm  D) Bellman-Ford Algorithm

**Answer: B**

**Q12. What is the primary objective of a Reverse Proxy like NGINX in web infrastructure?**

A) Filter outgoing corporate internet browsing requests  B) Distribute incoming client requests, terminate SSL, and cache static assets
C) Execute complex relational database queries in memory  D) Generate client-side JavaScript bundles dynamically

**Answer: B**

**Q13. Which memory region in a program's runtime memory is managed dynamically via malloc/free or new/delete?**

A) Call Stack  B) Heap
C) Data Segment  D) Text/Code Segment

**Answer: B**

**Q14. In public-key cryptography (e.g. RSA), which key is used by the recipient to decrypt a confidential message?**

A) Sender's Public Key  B) Sender's Private Key
C) Recipient's Public Key  D) Recipient's Private Key

**Answer: D**

**Q15. What is the purpose of an Index in a relational SQL database?**

A) Enforce foreign key constraints across tables  B) Accelerate data retrieval at the cost of additional storage and write overhead
C) Automatically compress old archival records  D) Encrypt sensitive tabular columns at rest

**Answer: B**

**Q16. Which principle of object-oriented design states that software entities should be open for extension but closed for modification?**

A) Single Responsibility Principle  B) Open/Closed Principle
C) Liskov Substitution Principle  D) Dependency Inversion Principle

**Answer: B**

**Q17. What is the time complexity to find an element in a balanced Binary Search Tree containing n nodes?**

A) O(1)  B) O(log n)
C) O(n)  D) O(n log n)

**Answer: B**

**Q18. In TCP/IP networking, what mechanism is used to establish a reliable connection between client and server?**

A) Two-way handshake  B) Three-way handshake (SYN, SYN-ACK, ACK)
C) Four-way tear down  D) UDP broadcast strobe

**Answer: B**

**Q19. What does the CAP theorem state regarding distributed data stores during network partitions?**

A) A system can achieve both Consistency and High Availability simultaneously without sacrifice  B) A system must trade off between C
C) Distributed latency is inversely proportional to cluster bandwidth  D) Partitioning causes immediate data loss in all replica nodes

**Answer: B**

**Q20. Which hashing algorithm generates a 256-bit fixed-length cryptographic digest?**

A) MD5  B) SHA-1
C) SHA-256  D) CRC32

**Answer: C**

**Q21. In Git version control, which command creates and switches to a new branch in a single step?**

A) git branch -d new-feature  B) git checkout -b new-feature
C) git merge --squash new-feature  D) git rebase new-feature

**Answer: B**

**Q22. What is the primary role of Docker containers in modern deployment pipelines?**

A) Emulate entire guest operating systems and virtual hardware  B) Package code and its dependencies into lightweight, isolated OS-le
C) Provide physical CPU virtualization using hypervisor BIOS flags  D) Replace the host OS kernel with a microkernel

**Answer: B**

**Q23. Which programming language paradigm treats computation as the evaluation of mathematical functions avoiding mutable state?**

A) Procedural Programming  B) Imperative Programming
C) Functional Programming  D) Object-Oriented Programming

**Answer: C**

**Q24. What is the purpose of the Domain Name System (DNS) on the internet?**

A) Encrypt packet payloads across wireless routers  B) Translate human-friendly domain names to numerical IP addresses
C) Synchronize local machine clocks with atomic time standards  D) Assign MAC addresses to Ethernet network cards

**Answer: B**

**Q25. What is a major advantage of WebAssembly (Wasm) in modern browsers?**

A) Completely replaces CSS for styling webpages  B) Executes compact binary code at near-native execution speed
C) Bypasses browser cross-origin security sandboxes  D) Provides native direct access to system disk drives without permission

**Answer: B**

**Q26. Which data structure is fundamentally utilized to implement recursive function call frames in computers?**

A) Call Stack  B) FIFO Queue
C) Priority Queue  D) Adjacency Matrix

**Answer: A**

**Q27. What is the mathematical definition of a Prime Number?**

A) Any number that cannot be divided by 2  B) A positive integer greater than 1 having exactly two positive divisors: 1 and itself
C) Any non-negative integer resulting from the square of a natural number  D) Any real number that can be expressed as a ratio of two

**Answer: B**

**Q28. Which fundamental quantum computing unit replaces the classical binary bit?**

A) Trit  B) Qubit (Quantum Bit)
C) Femtobit  D) Nanobit

**Answer: B**

**Q29. What is the primary role of a Content Delivery Network (CDN)?**

A) Perform transactional database rollbacks  B) Cache web assets geographically closer to users to reduce latency
C) Compile TypeScript source files into native machine code  D) Manage Kubernetes cluster node autoscaling

**Answer: B**

**Q30. In machine learning, what term describes a model learning noise and memorizing training data rather than generalizing?**

A) Underfitting  B) Overfitting
C) Regularization  D) Gradient Descent

**Answer: B**